



Question 2: Library Inventory Management (Pandas)

1. DataFrame Creation

```
In [8]: import pandas as pd

data = {
    "Key": ["Book1", "Book2", "Book3", "Book4", "Book5"],
    "Title": [
        "The Great Gatsby",
        "To Kill a Mockingbird",
        "1984",
        "Pride and Prejudice",
        "The Catcher in the Rye"
    ],
    "Author": [
        "F. Scott Fitzgerald",
        "Harper Lee",
        "George Orwell",
        "Jane Austen",
        "J.D. Salinger"
    ],
    "Year": [1925, 1960, 1949, 1813, 1951],
    "Availability": [True, False, True, True, True]
}

library_df = pd.DataFrame(data)
library_df.set_index("Key", inplace=True)
library_df
```

```
Out[8]:
```

	Title	Author	Year	Availability
Key				
Book1	The Great Gatsby	F. Scott Fitzgerald	1925	True
Book2	To Kill a Mockingbird	Harper Lee	1960	False
Book3	1984	George Orwell	1949	True
Book4	Pride and Prejudice	Jane Austen	1813	True
Book5	The Catcher in the Rye	J.D. Salinger	1951	True

2. Updating Data

```
In [5]: library_df.loc["Book2", "Availability"] = True
```

```
library_df
```

```
Out[5]:
```

	Title	Author	Year	Availability
Key				
Book1	The Great Gatsby	F. Scott Fitzgerald	1925	True
Book2	To Kill a Mockingbird	Harper Lee	1960	True
Book3	1984	George Orwell	1949	True
Book4	Pride and Prejudice	Jane Austen	1813	True
Book5	The Catcher in the Rye	J.D. Salinger	1951	True

3. Search & Selection

```
In [6]: fitzgerald_books = library_df[library_df["Author"] == "F. Scott Fitzgerald"]
fitzgerald_books
```

```
Out[6]:
```

	Title	Author	Year	Availability
Key				
Book1	The Great Gatsby	F. Scott Fitzgerald	1925	True

```
In [7]: available_books = library_df[library_df["Availability"] == True]
available_books
```

```
Out[7]:
```

	Title	Author	Year	Availability
Key				
Book1	The Great Gatsby	F. Scott Fitzgerald	1925	True
Book2	To Kill a Mockingbird	Harper Lee	1960	True
Book3	1984	George Orwell	1949	True
Book4	Pride and Prejudice	Jane Austen	1813	True
Book5	The Catcher in the Rye	J.D. Salinger	1951	True

```
In [ ]:
```